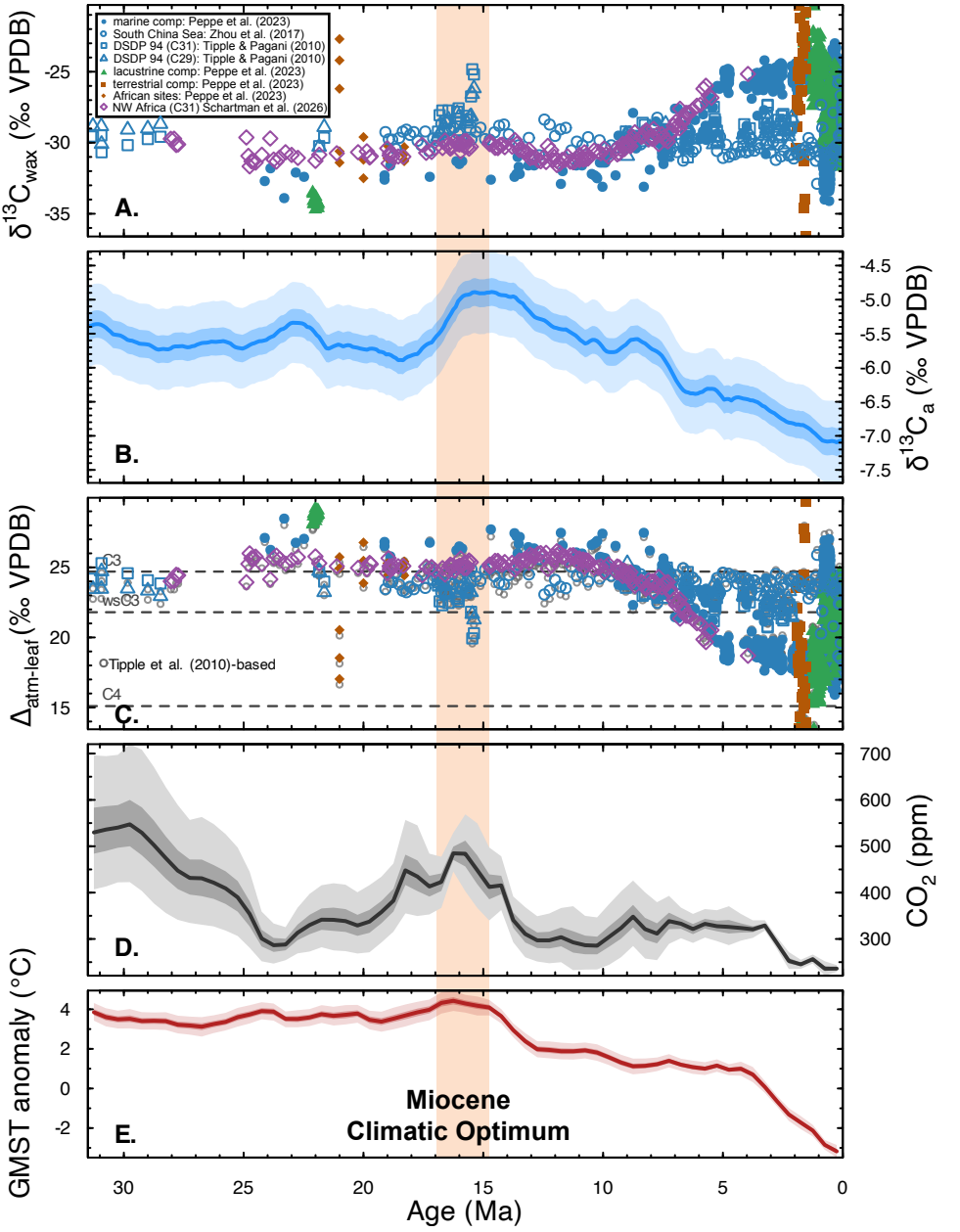

marine comp: Peppe et al. (2023)
South China Sea: Zhou et al. (2017)
DSDP 94 (C31): Tipple & Pagani (2010)
DSDP 94 (C29): Tipple & Pagani (2010)
lacustrine comp: Peppe et al. (2023)
terrestrial comp: Peppe et al. (2023)
African sites: Peppe et al. (2023)
NW Africa (C31) Schartman et al. (2026)
δ13Cwax (‰ VPDB)
-25
-30
-35
A.
B.
-4.5
-5.0
-5.5
-6.0
-6.5
-7.0
-7.5
δ13Ca (‰ VPDB)
Δatm-leaf (‰ VPDB)
25
20
15
C3
ws C3
Tipple et al. (2010)-based
C4
C.
D.
700
600
500
400
300
CO2 (ppm)
GMST anomaly (°C)
4
2
0
-2
E.
Miocene
Climatic Optimum
30
25
20
15
10
5
0
Age (Ma)